

Power-Tower-Based Current Control of Boost Converters with a Bounded Sliding Surface

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Abstract—Robust and efficient control of DC–DC boost converters is crucial for applications subject to severe load variations and reference changes. The Super-Twisting (ST) algorithm theoretically provides a robust and continuous control law; however, its convergence speed and robustness strongly depend on the choice of constant gains. This paper proposes a Power-Tower Super-Twisting (PTST) controller and a modified variant for use in the current control loop of a DC–DC boost converter. The proposed approach replaces the sign function in the classical ST algorithm with a power-tower function, introducing a state-dependent gain that accelerates convergence when the tracking error is large. The modified PTST, based on a normalized sliding surface, further bounds this gain to prevent numerical overflow and reduce overshoot while preserving finite-time stability. Simulation results under reference change and load disturbance scenarios confirm that the proposed controllers achieve faster convergence than the conventional ST controller.

Index Terms—power tower function, super-twisting, boost converter, model uncertainties, robust control, sliding mode control

I. INTRODUCTION

DC–DC boost converters are widely used in renewable energy systems, electric vehicles, and related applications where stable voltage and current regulation must be maintained despite load variations, input disturbances, and parameter uncertainties. These requirements motivate the development of robust control strategies.

Sliding mode control (SMC) is a well-established robust nonlinear control technique capable of handling uncertainties and disturbances [1], [2]. It has been successfully applied to power systems [3], [4] and electromechanical systems [5]. SMC forces system trajectories onto a predefined sliding surface using a discontinuous (or set-valued) control law, ensuring robustness.

Classical first-order SMC requires high constant gains to guarantee robustness and fast convergence under worst-case disturbances. However, these gains intensify the well-known chattering phenomenon caused by control discontinuities. To reduce chattering, the Super-Twisting (ST) algorithm [6] generates a continuous control signal while preserving finite-time convergence of both the sliding variable and its derivative. Nevertheless, ST performance remains strongly dependent on constant gain selection: large gains improve convergence and disturbance rejection but may also induce actuator saturation, numerical overflow, amplified transients, and chattering in digital implementations.

To address this trade-off, adaptive and variable-gain SMC strategies have been proposed. Adaptive approaches [7], [8] adjust gains online according to the sliding variable, improving flexibility compared with constant-gain ST controllers. However, they introduce additional adaptation dynamics and tuning parameters. In [9], several gains are linked through a single adaptive law, reducing tuning complexity but potentially constraining performance.

Variable-gain approaches provide state-dependent amplification without additional adaptation laws. In [10], perturbation-bound-dependent gain selection was proposed for the ST algorithm, while a Sugeno-type fuzzy controller was employed in [11] to mitigate chattering. Barrier-function-based methods [12], [13] also reduce control gains near the sliding surface to alleviate chattering. However, gain limitations are not explicitly addressed in these methods.

The Power-Tower-Function (PTF), introduced in [14], is a variable-gain switching function whose amplification increases with the sliding variable magnitude. The terminology “Power Tower” is inspired by the notion of tetration, i.e., repeated or nested exponentiation. This property makes the PTF attractive for accelerating convergence. In this work, a PTF-based Super-Twisting (PTST) controller and a modified PTST (MPTST) controller are proposed for inner-loop current control of DC–DC converters. The conventional sign function of the ST algorithm is replaced by a PTF, which amplifies the control effort far from the sliding surface while behaving similarly to the sign function near zero. However, recursive exponentiation may generate excessively large gains, potentially causing overshoot, numerical overflow, or actuator saturation.

Additional discussion regarding overshoot and saturation effects in the ST integral term can be found in [15]. A common mitigation strategy consists in scaling down the PTF argument, but this enlarges the region where amplification remains close to unity, thereby reducing the improvement compared with classical ST. To overcome this limitation, the proposed PTST with modified sliding surface introduces a bounded PTF formulation that limits gain growth without down-scaling, preserving fast convergence while preventing numerical overflow or/and actuator saturation.

Accordingly, the main contributions of this paper are: label=•

- The development of a **PTST** controller by replacing the sign function in the classical ST algorithm with a PTF

to enhance convergence speed.

- The introduction of a **Modified PTST (MPTST)** controller that bounds gain growth and reduces overshoot and actuator saturation.
- A **finite-time stability proof** for both controllers using Lyapunov analysis.
- The **first application of the PTF to the control of a physically motivated dynamical system**.
- A **comparative study** between ST, PTST, and MPTST controllers applied to a boost converter, demonstrating improved convergence and reliability.

The remainder of the paper is organized as follows. Section II presents the boost-converter model and preliminaries. Section III introduces the PTST controller, while Section IV presents the modified PTST approach. Section V validates the proposed methods through two case studies, and Section VI concludes the paper.

II. PRELIMINARIES

A. Dynamical Model of Boost Converter

The state equations of the boost converter operating in continuous-conduction mode (Figure 1) are:

$$\begin{aligned} \dot{v}_C &= \frac{1}{C}(ui_L - \frac{v_C}{R}) \\ \dot{i}_L &= \frac{1}{L}(-uv_C + v_{in}) \end{aligned} \quad (1)$$

where i_L and v_C are the inductor current and capacitor voltage, respectively; $u = 1 - D$ with D the duty cycle; v_{in} is the input voltage; and L , C , and R denote the inductance, capacitance, and load resistance.

Assumption 1: The output voltage is assumed to be regulated either by the DC bus or by an outer-loop controller in a cascaded structure.

Under *Assumption 1*, Tikhonov's theorem [16] allows decomposition of (1) into a fast inner current loop and a slow outer voltage loop, enabling independent controller design. The current-error dynamics are:

$$\dot{\varepsilon} = \frac{1}{L}(-uv_C + v_{in}) + \delta \quad (2)$$

where $\varepsilon = i_L - i_L^*$ and δ represents disturbances of the current dynamics.

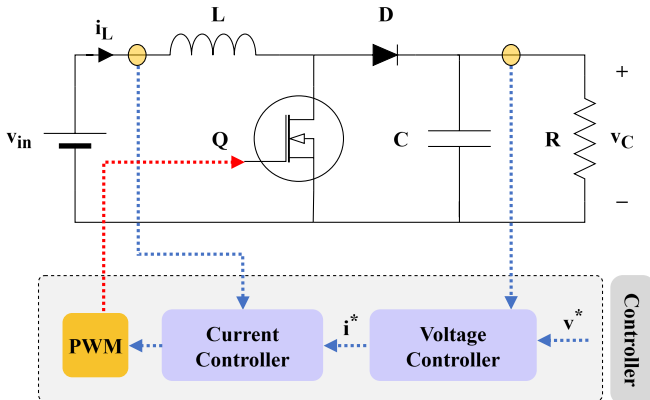


Fig. 1. Schematic of the cascaded loop controller for a boost converter.

Assumption 2: The disturbance δ in (2) is at least C^1 and globally Lipschitz:

$$\exists \Gamma > 0 \text{ s.t. } \forall t, |\dot{\delta}(t)| < \Gamma. \quad (3)$$

An averaged converter model is considered; hence, measurements are assumed filtered, which is reasonable with respect to the converter switching frequency.

B. Power-Tower Function

This section recalls the Power-Tower Function (PTF) introduced in [14], used later in the controller design.

Definition 1: $\text{PTF}_\beta(\cdot)$ is defined as:

$$\text{PTF}_\beta(z) \in |z|^{\beta} \text{sgn}(z) \quad (4)$$

where β controls the PTF growth rate. Larger β values increase amplification far from the origin.

Lemma 1 ([14]): For $z \neq 0$, the PTF derivative is:

$$\frac{d|z|^{\beta} \text{sgn}(z)}{dt} = |z|^{\beta-1} (1 + \beta \ln(|z|)) \frac{dz}{dt} \quad (5)$$

and for $z \neq 0$, $\beta > 0$:

$$\min(|\text{PTF}_\beta(z)|) = e^{\frac{-1}{\beta}}. \quad (6)$$

Lemma 2 ((Finite-time Stability [17])): Consider:

$$\dot{x} \in f(x, t) \quad (7)$$

where $f : \mathbf{R}^n \rightarrow \mathbf{R}^n$ and $0 \in f(0, t)$ for all t . Let $V(x)$ be positive definite and proper. System (7) is finite-time stable if:

$$\exists V(x) > 0 : \forall x \in X : \dot{V}(x) = \nabla^T V \cdot f(x, t) < -kV^a \quad (8)$$

with $k > 0$ and $a \in (0, 1)$. The settling time satisfies:

$$T \leq T_{\max} = \frac{V^{1-a}(0)}{k(1-a)}. \quad (9)$$

Lemma 3 ((Corollary of the min-max theorem [18])):

For a Hermitian matrix A , the Rayleigh quotient satisfies:

$$\lambda_{\min}(A) \leq \frac{x^T A x}{\|x\|^2} \leq \lambda_{\max}(A), \quad \forall x \neq 0 \quad (10)$$

where $\|\cdot\|$ denotes the Euclidean norm.

III. PTST CONTROLLER DESIGN

Finite-time stability of (2) under the proposed PTST controller is established below. The control input is decomposed as:

$$u = u_{eq} + u_{sw}. \quad (11)$$

The equivalent control maintaining the sliding surface is:

$$u_{eq} = \frac{v_{in}}{v_C}. \quad (12)$$

Theorem 1: The PTST controller:

$$\begin{aligned} u_{sw} &= \frac{L}{v_C} (k_1 |s|^{1/2} \text{PTF}_\beta(s) + \omega) \\ \dot{\omega} &\in -k_2 \text{PTF}_\beta(s) \end{aligned} \quad (13)$$

with $s = \varepsilon$, guarantees finite-time convergence of (2) for bounded initial conditions and properly chosen k_1, k_2 .

Proof: Substituting (11)–(13) into (2) gives:

$$\begin{aligned}\dot{s} &= -k_1 |s|^{1/2} \text{PTF}_\beta(s) + \omega + \delta \\ \dot{\omega} &\in -k_2 \text{PTF}_\beta(s).\end{aligned}\quad (14)$$

Defining $\rho = \omega + \delta$:

$$\begin{aligned}\dot{s} &= -k_1 |s|^{1/2} \text{PTF}_\beta(s) + \rho \\ \dot{\rho} &\in -k_2 \text{PTF}_\beta(s) + \dot{\delta}.\end{aligned}\quad (15)$$

Let:

$$\xi = [\xi_1 \quad \xi_2]^T, \quad \xi_1 = |s|^{1/2} \text{sgn}(s), \quad \xi_2 = \rho. \quad (16)$$

Then:

$$\dot{\xi} = |\xi_1|^{-1} A \xi \quad (17)$$

with

$$A = \frac{1}{2} \begin{bmatrix} -k_1 |\text{PTF}_\beta(s)| & 1 \\ -2(k_2 |\text{PTF}_\beta(s)| + \text{sgn}(s)\dot{\delta}) & 0 \end{bmatrix}. \quad (18)$$

Consider the Lyapunov candidate function:

$$V = \xi^T P \xi. \quad (19)$$

There exists a positive definite matrix P such that it satisfies $(A^T P + P A) = -Q < 0$. The conditions ensuring existence of $P > 0$ are given in [Appendix I](#). Hence:

$$\dot{V} = -|\xi_1|^{-1} \xi^T Q \xi. \quad (20)$$

Using [Lemma 3](#):

$$\|\xi\|^2 \leq \lambda_{\min}^{-1}(P) V \quad (21a)$$

$$\|\xi\|^2 \geq \lambda_{\max}^{-1}(P) V. \quad (21b)$$

Since $|\xi_1| \leq \|\xi\|$:

$$-|\xi_1|^{-1} \leq -\lambda_{\min}^{1/2}(P) V^{-1/2}. \quad (22)$$

Also:

$$\lambda_{\min}(Q) \|\xi\|^2 \leq \xi^T Q \xi. \quad (23)$$

Substituting (21b) into (23):

$$\lambda_{\min}(Q) \lambda_{\max}^{-1}(P) V \leq \xi^T Q \xi. \quad (24)$$

Using (22) and (24) in (20) yields:

$$\dot{V} \leq -\mu V^{1/2} \quad (25)$$

where

$$\mu = \lambda_{\min}^{1/2}(P) \lambda_{\min}(Q) \lambda_{\max}^{-1}(P) > 0.$$

By [Lemma 2](#), finite-time convergence is guaranteed with:

$$T_{\max} = \frac{2}{\mu} V_0^{1/2}, \quad V_0 = V(0, s(0), \rho(0)). \quad (26)$$

IV. PTST WITH MODIFIED SLIDING SURFACE

Although PTST achieves fast convergence, the PTF may produce excessively large gains causing overflow or actuator saturation. Downscaling the sliding variable alleviates this issue but weakens amplification over a wider operating range. To overcome this limitation, a bounded sliding surface is introduced:

$$s = \varphi_\kappa(\varepsilon) := \kappa \tanh(\varepsilon \tanh^{-1}(\kappa^{-1})) \quad (27)$$

where $\kappa > 1$.

The evolution of $\varphi_\kappa(\varepsilon)$ is shown in [Figure 2](#). The bounded behavior of $\varphi_\kappa(\varepsilon)$ limits the control effort and accounts for actuator constraints.

The function satisfies:

- $\varphi_\kappa(0) = 0$,
- φ_κ is strictly increasing,
- $\lim_{\varepsilon \rightarrow \infty} \varphi_\kappa(\varepsilon) = \kappa$.

Its derivative is:

$$\begin{aligned}\varrho(\varepsilon) &:= \frac{d\varphi_\kappa(\varepsilon)}{d\varepsilon} \\ &= \kappa \tanh^{-1}(\kappa^{-1}) \text{sech}^2(\varepsilon \tanh^{-1}(\kappa^{-1})).\end{aligned}\quad (28)$$

Since $\kappa > 1$, $\varrho(\varepsilon)$ is positive and bounded.

Remark 1: Compared with PTST, the MPTST introduces only the bounded positive factor $\varrho(\varepsilon)$, preserving the closed-loop structure. Therefore, the finite-time Lyapunov analysis of PTST directly extends to MPTST.

Remark 2: Because $\varphi_\kappa(\cdot)$ is strictly increasing and $\varphi_\kappa(0) = 0$, finite-time convergence of s implies finite-time convergence of ε .

Finally, to ensure practical applicability of the proposed MPTST controller, it is desirable to calculate the specific κ to enforce an arbitrary bound $\pm\eta$ on the PTF output. The asymptotic value of $\text{PTF}_\beta(\varphi_\kappa(\varepsilon))$ is given by:

$$\lim_{\varepsilon \rightarrow \infty} \text{PTF}_\beta(\varphi_\kappa(\varepsilon)) = \kappa^{\kappa^\beta} = \eta. \quad (29)$$

Taking logarithms:

$$\kappa^\beta \ln(\kappa) = \ln(\eta). \quad (30)$$

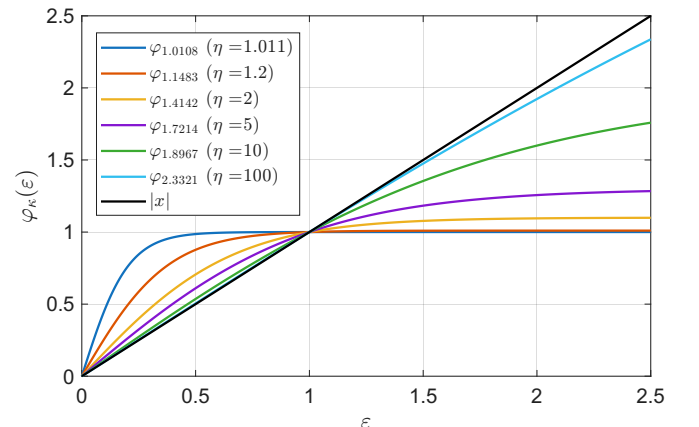


Fig. 2. Evolution of $\varphi_\kappa(x)$ for varying κ .

Defining $\lambda := \ln(\kappa)$:

$$\beta \lambda e^{\beta \lambda} = \beta \ln(\eta). \quad (31)$$

Using the Lambert W function¹:

$$\beta \lambda = W(\beta \ln(\eta)). \quad (32)$$

Hence:

$$\kappa = e^{\left(\frac{1}{\beta} W(\beta \ln(\eta))\right)} = \left(\frac{\beta \ln(\eta)}{W(\beta \ln(\eta))}\right)^{1/\beta}. \quad (33)$$

The left side of Figure 3 shows the minimum predicted by Lemma 1, while the right side illustrates the bounded behavior of $\text{PTF}_\beta(\varphi_\kappa(\varepsilon))$ due to the bounded sliding surface.

V. SIMULATION RESULTS

The proposed PTST control strategies are validated via MATLAB[®]/Simulink[®] simulations and compared with the classical ST controller for boost converter current control. In many practical systems, particularly in smart power grids using Buck, Boost, or SEPIC converters, the current reference is provided by a supervisory controller [20]. Accordingly, here a cascaded control structure is utilized, where a ST-based controller regulates the inner current loop and an outer integral proportional (IP) voltage controller generates the current reference under a singular perturbation framework.

The converter parameters are: inductance $L = 2.2$ mH, capacitance $C = 500$ μ F, load resistance $R = 100$ Ω , and nominal input voltage $v_{\text{in}} = 40$ V. The control gains are chosen as follows: for the voltage loop, $k_P = 0.141$, $k_I = 10$; and for the current loop, $k_1 = 500$, $k_2 = 15.7 \times 10^4$, $\beta = 2$, and $\eta = 10$. Moreover, an ode1 solver with a step size of $2.5\text{e-}7$ is used for the simulations.

A. Scenario 1: step change in reference voltage

A voltage reference step change from 200V to 300V is considered at $t = 0.1$ s in this scenario. The corresponding voltage response, presented in Figure 4, shows that the PTST controllers provide faster dynamics than the classical ST controller. As a result, the ST controller violates the singular perturbation assumptions, leading to oscillations around the reference value and preventing voltage convergence. Similar oscillations can be observed in the current waveforms

¹The Lambert W function satisfies $W(z)e^{W(z)} = z$ [19].

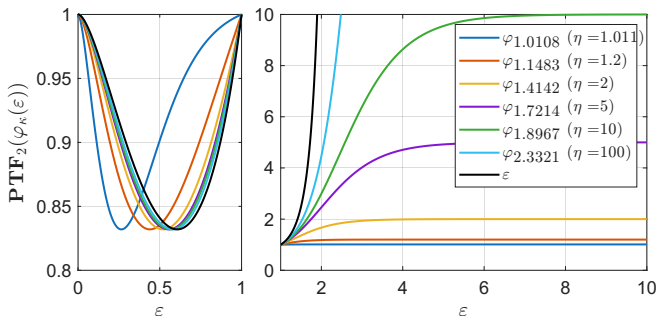


Fig. 3. Evolution of $\text{PTF}_\beta(\varphi_\kappa(\varepsilon))$ with $\beta = 2$ and varying κ . Left: $\varepsilon \in (0, 1]$. Right: $\varepsilon > 1$.

depicted in Figure 5 and Figure 6, as well as in the duty-cycle response in Figure 7. Moreover, comparison of the PTF functions of the PTST controllers with the sign function of the ST controller illustrates the extent of gain amplification provided by the PTST approaches in Figure 8.

B. Scenario 2: load variation

In this scenario, the controllers are investigated under a load variation. The load is changed from $R = 100$ to $R = 10$ at $t = 0.1$. As shown in Figure 9, the PTST controllers exhibit significantly better performance than the ST controller. The current responses in Figure 10 and the corresponding sliding variable evolutions in Figure 11 further demonstrate the faster convergence achieved by the PTST controllers. The key advantage of the MPTST is highlighted in figure Figure 13, where the PTST controller reaches a gain amplification of -2.56×10^{12} , while the MPTST constrains the gain to ± 10 . This bounded behavior prevents duty-cycle saturation in Figure 12, which is observed with the PTST.

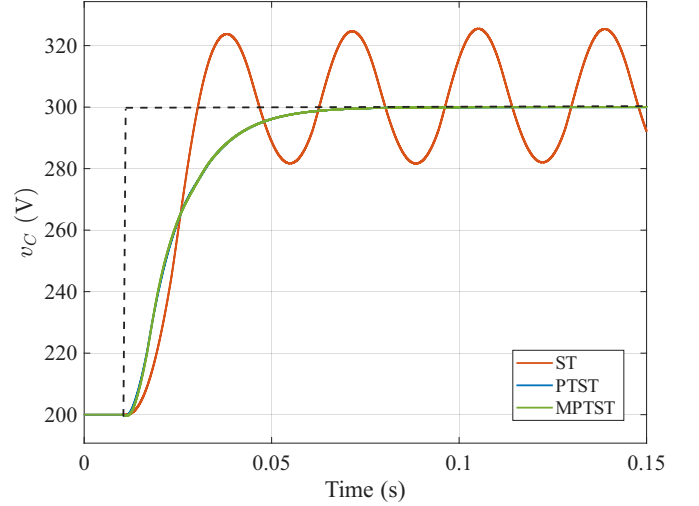


Fig. 4. Output voltage waveforms under a step change in the reference voltage.

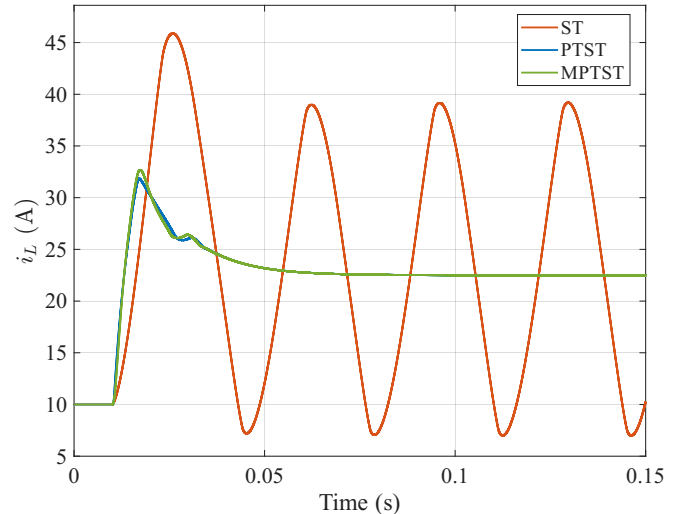


Fig. 5. Current waveforms under a step change in the reference voltage.

VI. CONCLUSION

In this paper, two novel PTST strategies are proposed for current control of a DC–DC boost converter. A stability analysis is performed, and finite-time convergence of the proposed methods is established. The main advantage of these controllers lies in their gain amplification effect, which enhances performance when the system operates far from the sliding surface. Moreover, the PTST with a bounded sliding surface prevents the ultra-high gains associated with the PTF. Simulation results show the superiority of the proposed bounded PTF controller compared to the unbounded one and the conventional constant-gain super-twisting.

APPENDIX I

COMPUTATION OF A FEASIBLE MATRIX P

Details regarding the computation of matrix P are provided in: https://github.com/naser-padar/Boost_MPTST

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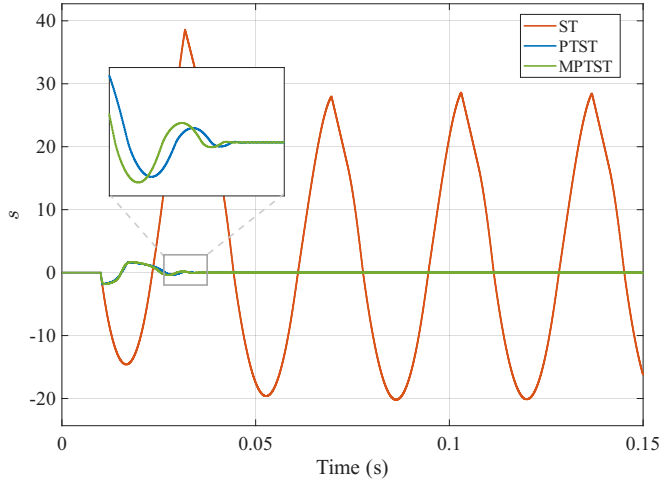


Fig. 6. Current error waveforms under a step change in the reference voltage.

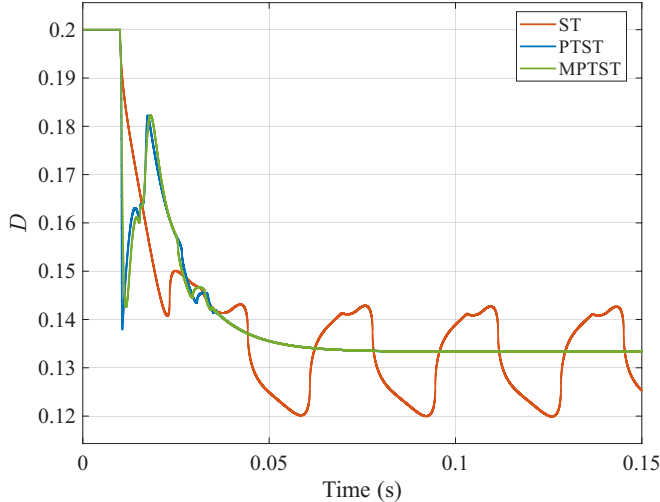


Fig. 7. Duty cycle waveforms under a step change in the reference voltage.

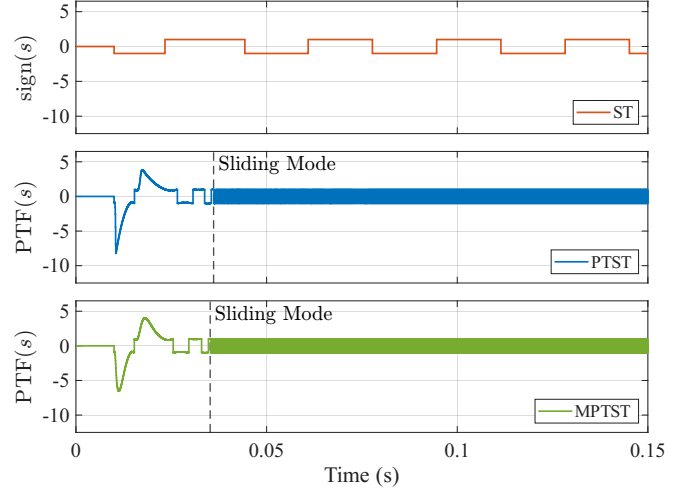


Fig. 8. Comparison of PTF and sign functions under a step change in the reference voltage.

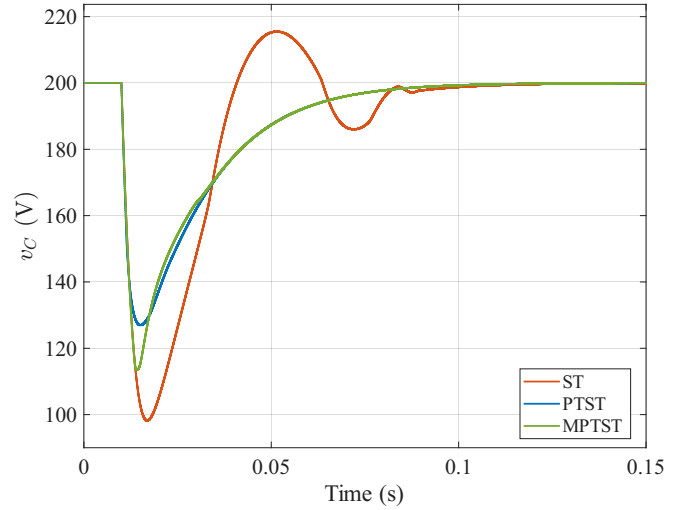


Fig. 9. Output voltage waveforms under load variation.

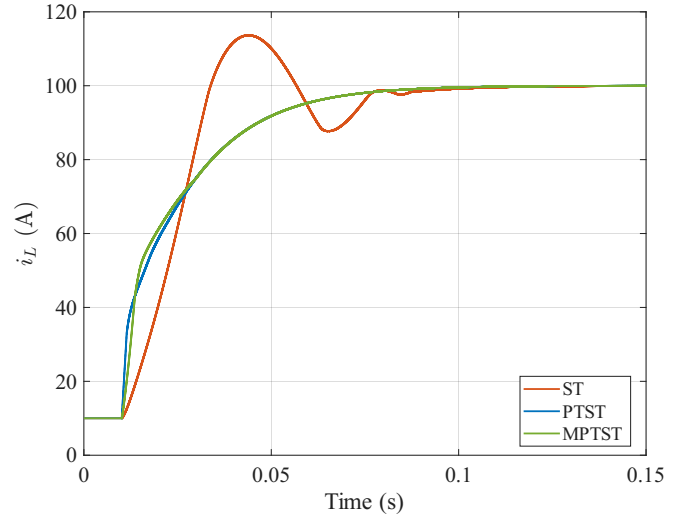


Fig. 10. Current waveforms under load variation.

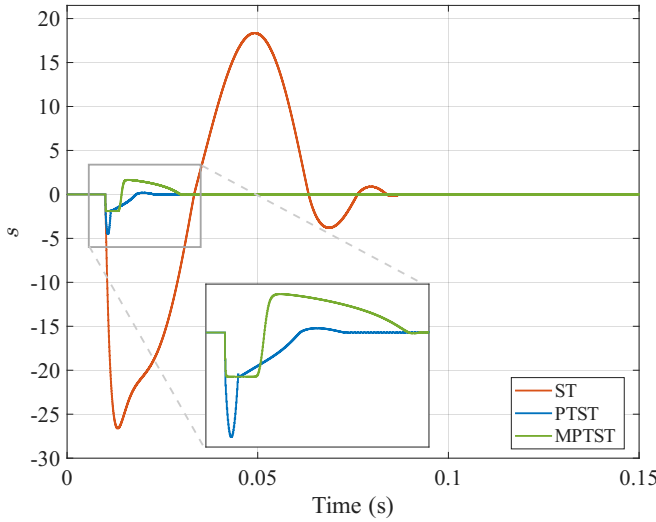


Fig. 11. Evolution of sliding variables under load variation.

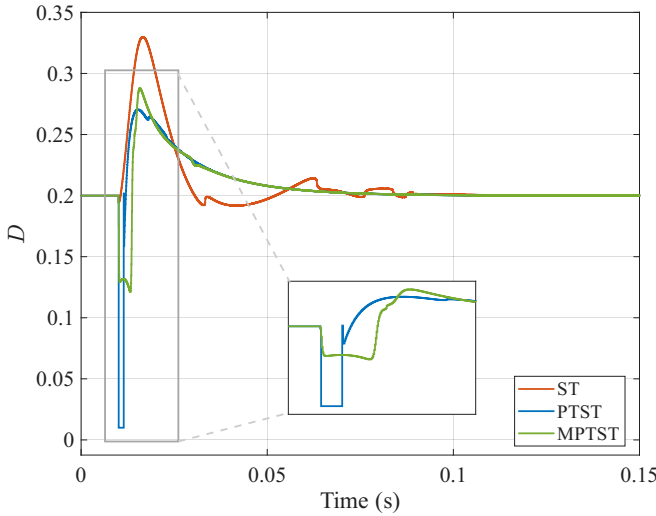


Fig. 12. Duty-cycle waveforms under load variation.

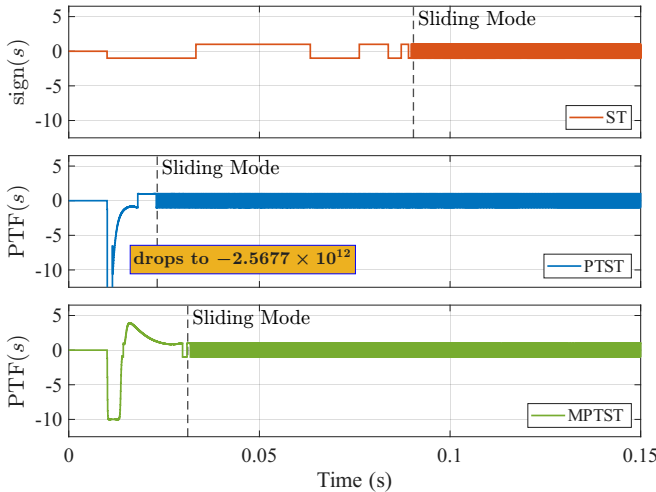


Fig. 13. Comparison of PTF and sign functions under load variation.

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